Event-Density Channels and the Global Geometry of Wave–Black-Hole Scattering

ED Interpretation of: "Gravitational Wave Scattering in Spinless WQFT"

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Abstract

A gravitational wave is a minimal ED-channel: a single participation rule with almost no internal ED-structure. A black hole is a maximal ED-structure: a fully individuated object whose steep ED-gradients define a global manifold. When such a minimal channel encounters such a maximal structure, the interaction is not a sum over histories. It is a global reorientation of the wave's participation rule inside the black hole's ED-channel.

The recent WQFT analysis of gravitational-wave scattering demonstrates that the S-matrix exponentiates in partial-wave space and maps directly onto the BHPT phase shift. In ED, exponentiation is the unmistakable signature of a single global ED-channel: the wave is not exploring many configurations; it is being carried along a unique structural pathway determined by the black hole's ED-gradients. The BHPT phase shift is the global invariant of this channel—the total reorientation accumulated as the wave traverses the ED-structure.

Helicity preservation and helicity flip arise as the two stable ways a minimal ED-channel can complete its participation rule inside a high-multiplicity gradient. What appear as "partial waves" in GR are, in ED, the stable alignment directions imposed by the black hole's geometry, not internal modes of the wave. The scattering process is therefore the simplest demonstration that ED-channels, not modal decomposition or quantum superposition, govern global structure.

This ED-I paper unifies the WQFT/BHPT result with the broader ED architecture developed in ED-I-013 and the Calabi–Yau ED-I-025 paper. In all cases—quantum algorithms, two-body scattering, and now wave–black-hole interactions—global invariants emerge whenever ED-gradients restrict the configuration space. The gravitational-wave phase shift is the global geometry of the ED-channel made observable.

1. Introduction: Minimal Channels in Maximal ED-Gradients

A gravitational wave scattering off a black hole is the cleanest possible demonstration of how a minimal ED-channel behaves when it enters a maximal ED-gradient. A gravitational wave carries almost no internal ED-structure: it is a single participation rule, a low-multiplicity channel, a thread with no capacity to encode orientation or geometry internally. A black hole, by contrast, is a fully individuated ED-object. Its ED-gradients are steep, its temporal tension is enormous, and its structure is globally coherent. When a minimal channel encounters such a maximal structure, the interaction is not a dynamical exchange of quanta. It is a global reorientation of the wave's participation rule inside the black hole's ED-channel.

The recent work by Bautista, Driesse, Haddad, and Jakobsen develops the worldline quantum field theory (WQFT) framework for this process and shows that the S-matrix exponentiates in partial-wave space. In standard language, this means the scattering amplitude reorganizes into a phase shift that matches black-hole perturbation theory (BHPT). In ED terms, exponentiation is the signature of a single global ED-channel: the wave is not exploring many paths; it is following one structural manifold determined by the black hole's ED-gradients.

This paper therefore provides a second, independent confirmation—alongside the Calabi–Yau three-fold result in two-body scattering—that high-precision gravitational calculations reveal the global geometry of ED-channels. Where the two-body problem exposed Calabi–Yau periods as global invariants of a radiative ED-channel, the single-body problem exposes the BHPT phase shift as the global invariant of a wave-scattering ED-channel. Both results arise from the same architectural principle: when ED-gradients restrict the configuration space, the system's observables inherit the global geometry of the ED-channel.

(Parenthetical cue: locally, the interaction forms a saddle—compression along the radial direction, expansion along the angular direction. Globally, this saddle extends into the full ED-channel that determines the phase shift.)

2. ED Ontology: The Structural Components Relevant to Wave Scattering

The ED interpretation of gravitational-wave scattering requires only the core elements of the ontology—minimal channels, high-multiplicity gradients, participation rules, and global ED-geometry. These components determine how a wave evolves when it encounters a black hole's ED-structure.

Minimal ED-Channels (the gravitational wave)

A gravitational wave is a low-multiplicity ED-channel. It carries a single participation rule with almost no internal ED-structure. It cannot encode continuous orientation internally; its helicity is a pre-geometric orientation rule, not a geometric vector. Minimal channels are fragile: they cannot resist external gradients and must rewrite their participation rule when forced by a high-multiplicity structure.

High-Multiplicity ED-Gradients (the black hole)

A black hole is a maximally developed ED-object. Its ED-gradients are steep, coherent, and globally structured. Any minimal channel entering this region is dominated by the black hole's ED-flow. The wave does not bring its own geometry into the interaction; it inherits the geometry of the black hole's ED-channel.

Participation Rules and Rewrite-on-Interaction

Because the wave is a minimal channel, its participation rule cannot remain unchanged. The black hole's ED-structure forces a rewrite-on-interaction, analogous to rewrite-on-measurement in ED-I-013 and ED-I-Spin. Helicity preservation or helicity flip are not quantum mechanical curiosities; they are the two stable ways a minimal channel can complete its participation rule inside a high-multiplicity gradient.

Global ED-Geometry

The black hole's ED-channel is not local. It is a global manifold that constrains the wave's entire evolution. The wave must follow this channel from asymptotic past to asymptotic future. The global invariant of this channel is the BHPT phase shift, which appears naturally when the S-matrix exponentiates. In ED, the phase shift is the global curvature of the ED-channel.

These components are sufficient to reinterpret the entire WQFT/BHPT matching result. The exponentiation of the S-matrix, the emergence of the phase shift, the helicity structure, and the partial-wave decomposition all follow from the same architectural fact: a minimal ED-channel is being carried through a maximal ED-gradient.

3. Gravitational Waves as Minimal ED-Channels

A gravitational wave is the simplest kind of ED-object the universe produces. It carries almost no internal ED-structure, no individuated identity, and no capacity to encode continuous orientation. In ED terms, it is a minimal-multiplicity channel: a single participation rule propagating through spacetime with no internal degrees

of freedom to stabilize or resist external gradients. Its “helicity” is not a geometric vector but a pre-geometric orientation rule, the earliest form of orientation available to a system that has not yet accumulated enough ED to encode geometry.

Because a gravitational wave is a minimal channel, it cannot maintain its participation rule when it encounters a high-multiplicity ED-structure. It has no internal ED-gradients to anchor its identity. Its orientation is not self-generated; it is inherited from the ED-environment. This is why gravitational waves are so sensitive to curvature: they do not carry their own geometry. They are carried by the geometry they traverse.

When such a wave approaches a black hole, the ED-gradients steepen dramatically. The wave’s participation rule is forced to align with the black hole’s ED-structure. Helicity preservation or helicity flip are not quantum mechanical curiosities; they are the two stable completion pathways available to a minimal channel being reoriented by a maximal ED-gradient. The wave does not “scatter” in the quantum sense. It completes its participation rule inside the black hole’s ED-channel.

The minimality of the wave is therefore not a limitation. It is the reason the scattering process reveals the ED-channel so cleanly. A minimal channel cannot hide the geometry of the structure it passes through. It expresses it.

4. Black Holes as Maximal ED-Structures

A black hole is the opposite extreme of ED-structure. It is a maximally individuated ED-object, a system whose internal ED-gradients are so dense and coherent that they define a global manifold. Its identity is fully developed. Its orientation is not a rule but a geometry. Its ED-structure is not local but global, extending from the horizon to asymptotic infinity through the curvature it imposes on the surrounding field.

In ED terms, a black hole is a high-multiplicity ED-structure. It possesses a vast number of stable participation pathways, all aligned by the global ED-flow generated by its mass. Any minimal channel entering this region is dominated by the black hole’s ED-gradients. The wave does not bring its own geometry into the interaction; it inherits the geometry of the black hole’s ED-channel.

This is why the scattering process is so rigid. The black hole’s ED-structure collapses the configuration space into a single global channel. The wave cannot explore arbitrary histories. It must follow the ED-channel determined by the black hole’s mass, curvature, and temporal tension. The global invariant of this channel is the BHPT phase shift, which appears naturally when the S-matrix exponentiates.

(Parenthetical cue: the steep ED-gradients form a local saddle, but the global ED-channel is the full manifold that the wave must traverse.)

The black hole’s ED-structure therefore plays two roles:

- It supplies orientation the wave cannot encode internally.
- It restricts the configuration space to a global channel whose geometry determines the scattering outcome.

This is why the WQFT amplitude exponentiates. Exponentiation is not a mathematical trick. It is the signature of a system evolving along a single ED-channel. The black hole’s ED-structure is so dominant that the wave’s evolution is globally constrained. The N-matrix is simply the generator of that constraint.

The scattering event is therefore not a dynamical exchange of gravitons. It is the global reorientation of a minimal ED-channel inside a maximal ED-gradient.

5. ED Interpretation of the WQFT Framework

Worldline quantum field theory (WQFT) is normally presented as a computational device: a way to treat a black hole as a point particle, expand the Einstein–Hilbert action, and compute scattering amplitudes using retarded propagators and worldline vertices. In ED, this entire framework has a clean structural meaning. WQFT is not summing over histories. It is encoding the participation geometry of a minimal ED-channel interacting with a high-multiplicity ED-structure.

A gravitational wave is represented in WQFT by a single graviton line: a minimal channel with one participation rule. The black hole's worldline is a high-multiplicity ED-structure whose gradients dominate the interaction. The retarded propagators enforce ED-causality: participation flows forward along the ED-channel and cannot loop back. The diagrammatic expansion is not a quantum superposition; it is a bookkeeping device for how the minimal channel is carried through the black hole's ED-gradients.

The "incoming plane wave" is simply the minimal channel before individuation. The "outgoing wave" is the same channel after completing its participation rule inside the black hole's ED-structure. The diagrams in WQFT encode the local deformations of the ED-channel, while the exponentiated S-matrix encodes the global geometry of the channel.

This is why WQFT matches BHPT so cleanly. Both are describing the same ED-channel: one in diagrammatic form, the other in differential-equation form. The agreement is not a coincidence. It is the structural inevitability of a minimal ED-channel being carried through a maximal ED-gradient.

6. Exponentiation of the S-Matrix as ED-Channel Inevitability

The most important conceptual result of the paper is that the S-matrix exponentiates in partial-wave space. In standard language, this means the scattering amplitude reorganizes into a phase shift that matches black-hole perturbation theory. In ED language, exponentiation is the unmistakable signature of a single global ED-channel.

A minimal ED-channel does not explore many paths. It does not interfere with itself. It does not sum over histories. It is forced to follow the unique structural manifold defined by the black hole's ED-gradients. The exponential form of the S-matrix,

$$S=e^{iN},$$

is simply the mathematical expression of this architectural fact. The operator N is the generator of the global ED-channel. It encodes the global curvature of the channel, not the local perturbations. The T-matrix elements correspond to local deformations of the ED-structure; the N-matrix corresponds to the global constraint imposed by the black hole.

This is the gravitational analogue of the global-participation phenomenon in ED-I-013. In the Deutsch oracle, the system does not evaluate many inputs. It applies one participation rule to a global ED-channel. In wave–black-hole scattering, the system does not explore many histories. It follows one ED-channel whose global invariant is the BHPT phase shift.

Exponentiation is therefore not a computational trick. It is the ED signature of a system whose evolution is globally constrained. The wave is not scattering in the quantum sense. It is completing its participation rule inside a global ED-channel. The phase shift is the global invariant of that channel.

7. ED Interpretation of the BHPT Phase Shift

The BHPT phase shift is normally introduced as a technical object: the infrared-finite quantity extracted from the asymptotic behavior of the Regge–Wheeler and Zerilli equations. In ED, the phase shift is not a computational artifact. It is the global invariant of the ED-channel that a gravitational wave must traverse when it interacts with a black hole.

A minimal ED-channel cannot carry its own geometry. It cannot maintain its own orientation. It cannot stabilize its participation rule against steep gradients. When such a channel enters the black hole's ED-structure, it is forced to follow the unique global manifold defined by the black hole's curvature and temporal tension. The wave's participation rule is reoriented continuously along this manifold. The total reorientation accumulated from past infinity to future infinity is the phase shift.

In ED terms:

- The local deformation of the wave's participation rule corresponds to the T-matrix.
- The global deformation corresponds to the N-matrix.
- The global invariant of the ED-channel is the BHPT phase shift.

This is why the phase shift is real, finite, and geometric. It is not a quantum interference effect. It is the global curvature of the ED-channel. The wave is not sampling many paths; it is being carried along one. The exponentiated S-matrix simply expresses this inevitability.

The WQFT result that the N-matrix maps directly to the BHPT phase shift is therefore not surprising in ED. Both objects encode the same structural fact: the wave's evolution is globally constrained by the black hole's ED-geometry. The phase shift is the ED-channel made observable.

8. Helicity Preservation, Helicity Flip, and Rewrite-on-Interaction

A gravitational wave can preserve or flip its helicity when scattering off a black hole. In standard GR, this is attributed to graviton self-interaction and the tensorial nature of the field. In ED, helicity behavior is not a dynamical curiosity. It is the structural consequence of a minimal ED-channel completing its participation rule inside a high-multiplicity ED-gradient.

Helicity is not a geometric vector. It is a pre-geometric orientation rule—the earliest form of orientation available to a system with almost no internal ED-structure. A minimal channel cannot store orientation internally. It inherits orientation from the ED-environment. When the wave enters the black hole's ED-structure, the steep gradients force a rewrite-on-interaction, analogous to rewrite-on-measurement in ED-I-013 and ED-I-Spin.

Two outcomes are possible:

- Helicity preservation: the wave's participation rule aligns with the ED-channel without contradiction.

- Helicity flip: the wave's participation rule is incompatible with the ED-channel and must rewrite to complete individuation.

These are not quantum choices. They are the two stable participation pathways available to a minimal channel being reoriented by a maximal ED-gradient.

This interpretation unifies several phenomena:

In ED-I-Spin, a low-multiplicity system cannot encode continuous orientation and must complete one of a few discrete pathways.

In ED-I-013, a minimal channel interacting with a high-multiplicity structure must rewrite its participation rule when incompatible.

In wave–black-hole scattering, a minimal channel must rewrite its helicity rule when the ED-channel demands it.

Helicity flip is therefore not a quantum mechanical oddity. It is the ED signature of a system too simple to carry its own orientation, being forced to complete its participation rule inside a structure far more developed than itself.

9. The ED-Channel as a Single Global Alignment Pathway

In ED, a gravitational wave does not decompose into "modes." It has no internal structure to support such a decomposition. It is a minimal ED-channel—a single participation rule carried through the black hole's ED-gradients. What physicists call "partial waves" is simply the mathematical projection of this single channel onto a basis. The ED ontology does not contain these basis elements. It contains only the global ED-channel and the stable ways a minimal rule can align with the black hole's structure.

A black hole supplies a complete, high-multiplicity ED-structure. Its curvature defines a global orientation field that any minimal channel must align with. The wave cannot choose among many internal configurations; it has only one participation rule. As it traverses the black hole's ED-channel, that rule is continuously reoriented by the surrounding gradients. The different "angular components" seen in BHPT are not internal modes of the wave. They are the different geometric directions along which the ED-channel constrains the wave's alignment.

The phase shift associated with each angular component is simply the amount of global reorientation the ED-channel imposes along that direction. The mathematics of spherical harmonics is a coordinate-dependent description of this geometry. The ED ontology contains only the geometry itself: a single global channel with multiple stable alignment directions determined by the black hole's ED-structure.

The scattering process is therefore not a sum over modes. It is a single ED-channel expressing its global geometry through the stable alignment directions available to a minimal ED-object.

10. Predictions and Consequences

The ED interpretation of wave–black-hole scattering predicts a series of structural consequences that extend beyond the specific calculations in the WQFT paper.

Higher-order PM predictions.

As the wave probes deeper into the black hole's ED-structure, the global ED-channel becomes more intricate. ED predicts that higher-order PM corrections will reveal additional global alignment constraints, not new internal modes. These constraints should appear as new geometric features in the phase shift, especially in the radiative sector.

Non-minimal operators and tidal effects.
The introduction of tidal operators in WQFT corresponds, in ED, to additional structural constraints on the ED-channel. These constraints modify the global geometry and therefore alter the wave's alignment pathway. ED predicts that tidal effects will manifest as changes in the global curvature of the ED-channel, not as new degrees of freedom of the wave.

Exponentiation as a structural diagnostic.
Exponentiation of the S-matrix indicates that the wave is following a single global ED-channel. ED predicts that exponentiation will fail only when the ED-channel becomes multi-branch, such as in strong tidal disruption, near-horizon mixing, or interactions with spinning black holes where the ED-channel twists. The breakdown of exponentiation would therefore signal a transition in the underlying ED-geometry.

Kerr scattering and channel twisting.
For rotating black holes, ED predicts that the ED-channel acquires a global twist due to frame dragging. This twist should produce helicity-dependent asymmetries and new geometric features in the phase shift. The WQFT framework is ideally suited to reveal these effects.

Unification with ED-I-013.
The exponentiation of the S-matrix and the emergence of the phase shift are gravitational analogues of the global-period extraction in Deutsch-style oracles. In both cases, the system is forced into a global ED-channel whose invariants appear directly in the observables. ED therefore predicts that any system—computational or gravitational—that restricts the configuration space through steep ED-gradients will reveal its global invariants in the same way.

The architectural conclusion is clear: wave–black-hole scattering is the simplest demonstration that ED-channels, not modal decomposition, govern global structure. The phase shift is the global invariant of the ED-channel, just as Calabi–Yau periods are the invariants of the two-body radiative channel and just as periods in Deutsch-style algorithms are the invariants of high-multiplicity participation channels.

References:

Bautista, Y. F., Driesse, M., Haddad, K., & Jakobsen, G. U. (2026). *Gravitational Wave Scattering in Spinless WQFT*. arXiv:2602.06125v2, 20 Feb 2026.